

Paint the World: A Sensor-Driven AR Painting Application

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Figure 1: An augmented photo exported from the Paint the World application, demonstrating some of the different brushes, tools, and materials available to the user.

KEYWORDS

augmented reality, geometry mesh generation, data processing

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1 INTRODUCTION

Augmented Reality (AR) has opened new opportunities for interactive digital art, where users can engage directly with their environment to create virtual compositions. Current AR and VR art applications, however, often restrict user interaction to simple tasks, lack the ability to generate procedural geometry at runtime, or require

external hardware. For example, Google's Tilt Brush necessitates the use of a VR headset, limiting accessibility and convenience.

Our project introduces an innovative AR painting application that leverages the full spatial capabilities of modern smartphones, offering an immersive painting experience without the need for additional hardware beyond an iPhone. By utilizing camera and orientation sensor data, the application precisely tracks the phone's position and orientation in 3D space. This advanced tracking capability allows users to "paint" digital strokes as if their device were a brush, creating a seamless and intuitive user experience.

This approach not only brings a new level of interactivity to digital art but also enhances creative freedom. Users can make dynamic changes in brushstrokes and effects based on their movement and input, offering a level of control and expressiveness that static digital art tools cannot match. The application supports a variety of brush settings, enabling artists to experiment with different stroke types, widths, and textures, all of which respond in real-time to the user's actions.

Moreover, the application's ability to generate procedural geometry at runtime sets it apart from many existing AR art tools. This feature allows for the creation of complex and intricate 3D structures, adding depth and dimension to the artwork. Users can sculpt and mold their digital strokes in a three-dimensional space, transforming their physical environment into a dynamic canvas.

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2 APPROACH

The core of our application lies in its ability to dynamically capture and interpret sensor data to render procedural geometry. We employ Apple's RealityKit geometry API to generate 3D digital paint strokes, anchored in AR, which users create by moving their devices through space. The application features multiple settings to modify the stroke characteristics:

- **Dynamic Width:** Users can alter stroke width by sliding their fingers across the screen while painting, adding a layer of depth and expression to their creations.
- **Brush Types:** Options include tube strokes to generate 3D stroke geometry, flat strokes for a more painterly effect, and ribbon strokes for elegant, twisting forms. There's also an option to create custom convex meshes by drawing the outline of a shape with your device.
- **Primitives:** Users can also place geometric shapes like boxes, spheres, and planes within their AR space, providing structural elements to their scenes.
- **Materials:** An in-app editor allows for real-time adjustments to the visual properties of strokes and shapes, such as color and reflectivity.
- **Geo Editing:** Post-placement adjustments can be made to objects' orientation, scale, and material properties, enhancing the compositional flexibility.

Additionally, users can capture and save their AR scenes to their camera roll or enhance them with VR backgrounds, such as skydomes, to create fully immersive environments.

3 IMPLEMENTATION AND RESULTS

Implementing this application required overcoming several technical challenges, particularly in data smoothing and real-time rendering. We developed a method to resample and smooth the logged sensor data based on user-defined stroke settings, ensuring fluid visual output regardless of hand jitter or rapid movements.

There are three primary data points that we capture each frame during brush stroke generation: device position relative to the AR anchor, device orientation, and the stroke width (based on the location of the touch interaction on the screen). The rate at which this data is collected is variable based on user-adjustable settings for resolution and smoothness.

The geometry generation pipeline begins with smoothing the collected data points in a multi-step process:

- (1) **Resample position data:** We optionally adjust the number of data points we want to use before we begin the smoothing process, and we normalize the spacing between all collected data points.
- (2) **Convert resampled position points to a bezier spline:** We use a bezier algorithm to create a smoothed path from the resampled positions, resulting in a smooth path with higher resolution at more acutely-angled sections of the path.
- (3) **Smooth radius and orientation data:** We use the collected radius and orientation data points to create a new mapped set of data that corresponds to each of the smoothed position points.

The smoothed values for position, orientation, and stroke width are then used to create a MeshBuffer for the 3D stroke geometry. This MeshBuffer is essential for defining the structure of the 3D stroke, which is then passed through the render pipeline for RealityKit to render over the AR scene. The generation of point positions, indices, and normals varies based on the user-adjustable brush settings, allowing for a high degree of customization and precision in the strokes.

Preliminary user feedback has been overwhelmingly positive, highlighting the ease of use and the novel experience of painting in augmented reality. Users appreciate the intuitive interface and the seamless integration of their physical movements into the digital strokes, making the painting process feel natural and immersive. Users have noted the application's potential not only for artistic expression but also for educational purposes and collaborative art projects.

4 CONCLUSION AND FUTURE WORK

Paint the World represents a significant advancement in digital art tools, offering artist interactivity and creative control within an augmented reality environment and moving toward democratizing this type of XR tool by making it more widely available through its minimal hardware requirements.

In its current state, Paint the World is fundamentally a proof-of-concept. Key features, such as the ability to save and load 3D scenes that users create, need to be implemented to transition the app into a viable market-ready product. Future development could focus on expanding the available toolset, including introducing various brush effects and more sophisticated AR capabilities like object recognition and interaction, which would significantly enhance the user experience.

Another exciting direction for future work includes exploring social features. Allowing users to share and collaborate on AR art pieces in real time would foster a community of digital artists and elevate the app's appeal. Additionally, integrating features that enable users to showcase their work in virtual galleries or participate in collaborative projects could further enrich the user experience.

This project not only broadens the scope of digital art but also showcases the transformative potential of AR technology. By turning ordinary environments into dynamic canvases for creativity, Paint the World shows how we interact with digital art and opens up new possibilities for artistic expression in everyday spaces.